

(Viscosity of Non-Newtonian Liquid)

Pre-Lab Plan

Unit Operations Lab 1

9/2/2025

Group 1, Tuesday, AM section

Team Member Names:

Joshua Olusoji

Omar Zaruk

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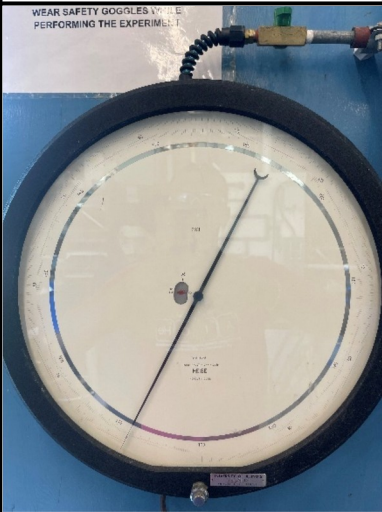
Qussay Suleiman

1. Experimental Objective (5 pts)

The objective of this experiment is to determine the non-newtonian characteristics of a 1wt% carbopol solution in water. We will use a capillary tube viscometer and vary the pressure, capillary tube length, and capillary tube orifice diameter to collect flowrate data and create an experimental plot to determine whether the solution is sheer-thinning or sheer-thickening.

2. Experimental

2.1 Apparatus (5 pts)

Equipment	Picture	Description
Pressure Regulator		A device used to adjust the air pressure delivered to the capillary tube viscometer. Regulates air pressure from the wall from 0-120 psia
Pressure Gauge		A device that measures and displays the pressure being delivered to the capillary tube viscometer.

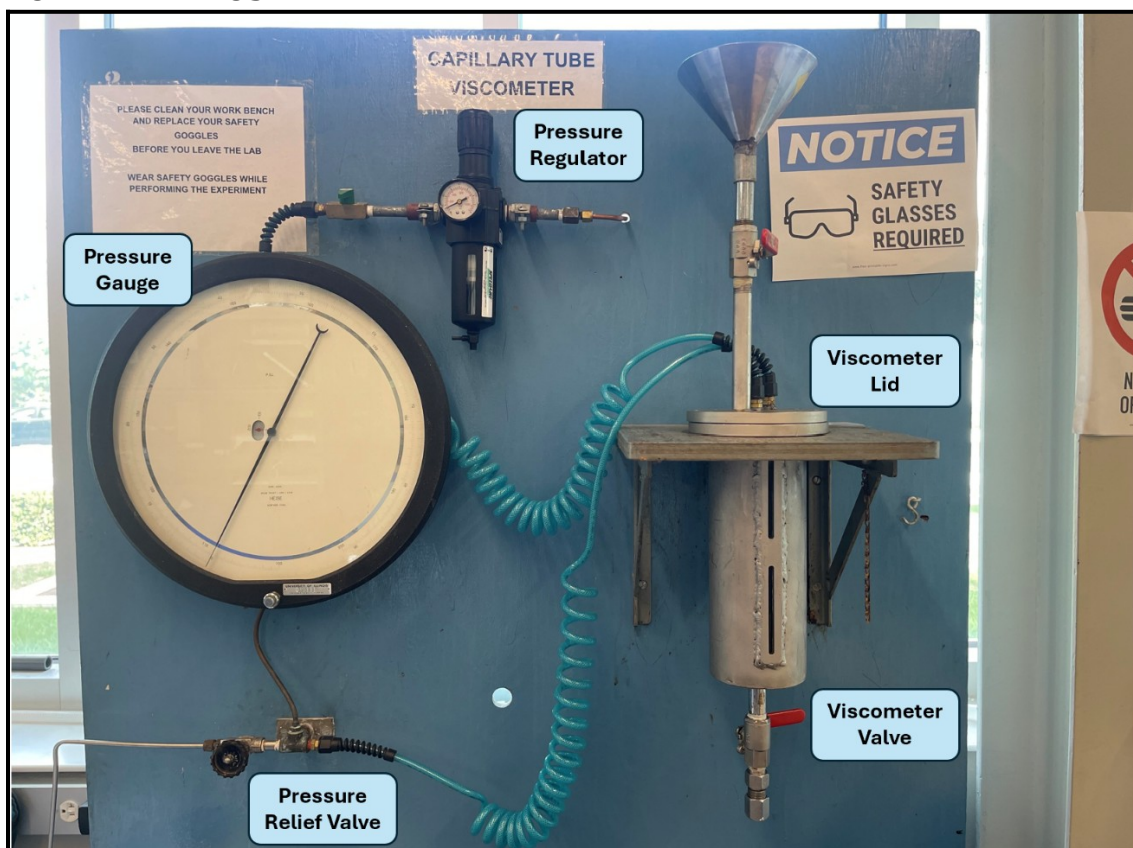
Pressure Relief Valve		A device used to relieve pressure from the system. Used as a safety measure in ensuring excessive pressure doesn't build up in the viscometer.
Viscometer Lid		The lid covers the viscometer and can be bolted down to be airtight.
Viscometer Valve		The valve allows fluid inside the viscometer to flow out of the viscometer, enabling data collection & viscosity calculations.

Capillary Tubes

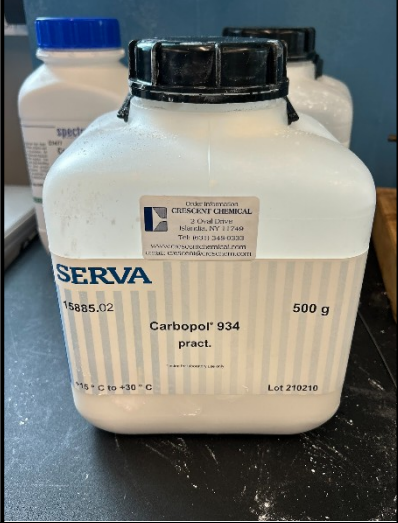


Piece of equipment that is attached to the exit of the viscometer. Different tubes offer varying length & orifice diameter, changing the shear experienced by the fluid inside the viscometer.

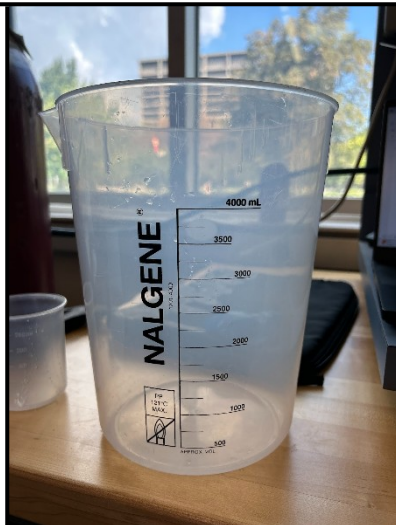
FULL APARATUS



2.2 Materials and Supplies (5 pts)

Materials/ supplies		Description
CARBOPOL 934		Polymer used to prepare the test solution (1.0 wt% in warm tap water)
Warm tap water		Solvent for dissolving CARBOPOL powder
1 N NaOH solution		Used to neutralize CARBOPOL solution to pH 5-7. Making sure it is honey like consistency

Beakers (1-4 L capacity)



Containers for mixing and preparing solutions

Stirring rods



For manual mixing support during solution preparation

Disposable gloves



For safe handling of chemicals
CARBOPOL powder, NaOH solution

Safety goggles		Eye protection during pressurization operations and when handling NaOH
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2.3 Experimental Plan (10 pts)

Solution Preparation

- 1) Add 2L of warm water and 1wt% Carbopol (20 g) into a beaker, then add the solution to a mixer and leave until the solution is fully dispersed with no clumps.
- 2) Slowly add and mix NaOH to the solution until neutralized to a pH of 5-7.

Experiment

- 1) Add the solution into the entrance of the vessel, apply grease to the joints, then tighten the screws to mount the lid.
- 2) Connect air line to the regulator and pressure gauge.
- 3) Choose a capillary tube (record its length L and inner diameter D). Install it securely
- 4) Set the regulator to the first ΔP setpoint, and allow 10s for steady flow.
- 5) Collect 50 mL effluent in a graduated cylinder for the specified time interval and record volume/mass
- 6) Record collection time and exact ΔP
- 7) Repeat for 2 additional ΔP setpoints.
- 8) Repeat the same ΔP setpoints with the remaining 2 capillary tubes
- 9) After all runs, flush with warm water

Variables

- Independent: ΔP (psig), capillary L and D (inches)
- Dependent: Volumetric flow rate Q (m^3/s), Shear Stress and rate, power law parameter m,n
- Controlled: Temperature, pH, composition, ambient pressure

Data and model correlation

- Compute $Q = V/t$
- Fit log-log relation from the laminar theory, $x = \log(Q/(\pi R^3))$, $y = \log((\Delta P)R/(2L))$
- Extract slope n, and intercept m
- Classification: $n < 1$ shear-thinning, $n = 1$ Newtonian, $n > 1$ shear-thickening

Project Safety Evaluation (10 pts)

	Yes/No	
Potential electrical hazards?	Yes	If yes, identify conditions: Bench power mixer near liquids; shock risk from wet cords/outlets. Keep cords and controller dry
Potential mechanical hazards?	Yes	If yes, identify conditions: Rotating mixer shaft (entanglement), glass capillary breakage/cuts. Keep hands and clothing clear.
Potential pressure hazards?	Yes	If yes, identify conditions: Air-pressurized vessel up to 60 psig ($\approx$ 75 psia). Risk of rupture/line blow-off/projectiles. Use rated vessel/tubing
Potential temperature hazards?	Yes	If yes, identify conditions: Warm tap water; exotherm when dissolving/adding NaOH to water and during neutralization. Add base slowly to water (never water to base)
Hazardous/toxic chemicals involved?	Yes	If yes, hazards involved: Sodium hydroxide (1 N): Corrosive—causes severe skin/eye burns; corrosive to metals. Handle as caustic. Carbopol/Carbomer 934 powder: Not GHS-classified as hazardous but dust may irritate eyes/airways/skin; avoid dust formation.
Gloves required?	Yes	If yes, specify type: Nitrile or neoprene gloves for NaOH work (splash protection)
MSDS reviewed by all team members?	Yes	List MSDS sheets reviewed: Sodium Hydroxide (1N)- safety sheet SERVA Carbopol 934— safety sheet
<i>Process Parameters</i>	<i>Max Expected</i>	<i>List location and possible hazards and precautions</i>
Temperature ($^{\circ}\text{C}$)	45 $^{\circ}\text{C}$	Where/when: During dissolution and NaOH neutralization. Hazards: Solution can heat from NaOH addition; splash/burn risk. Precautions: Add NaOH to water slowly, stir, allow cooling
Pressure (psia)	75 psia (60 psig)	Where/when: Pressurizing vessel for capillary runs. Hazards: Vessel/line failure, sudden release, capillary ejection. Precautions: Pressure regulator, relief valve to safe location. shield/face protection as needed
<i>Safety Controls</i>	<i>yes/no</i>	<i>If yes, Provide description</i>
<i>PPE for initial mixing:</i>	Yes	During the first wetting/dissolution of dry Carbopol, wear chemical splash goggles and mouth/nose protection to prevent eye irritation and inhalation of airborne powder
Pressure control valve	Yes	Use a pressure regulator/control valve on the air line set $\leq$ 60 psig, with a visible pressure gauge and a relief valve

I have read relevant background material for the Unit Operations Laboratory entitled: “*Non-Newtonian Viscosity*” and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance to standards and acceptable safety practices and will conduct all of my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

<u>Joshua Olsuji</u>	<u>9/2/2025</u>
signature of team member	date

<u>Omar Zaruk</u>	<u>9/2/2025</u>
signature of team member	date

<u>Sanad Abu Hweij</u>	<u>9/2/2025</u>
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<u>Qussay Suleiman</u>	<u>9/2/2025</u>
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